(a) Voxel-to-brick construction

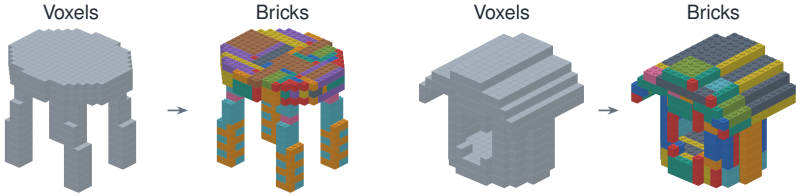
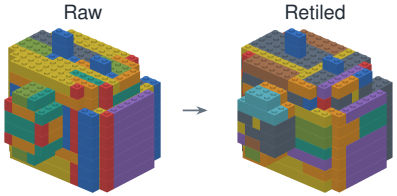


Table: 1,117 voxels → 197 bricks

Birdhouse: 937 voxels → 200 bricks

(b) Connectivity repair



Camera: raw → retiled

Brick type 1×1 1×2 1×3 1×4 1×6 1×8 2×2 2×3 2×4 2×6 2×8

Automated dataset construction
80k source shapes · 21 categories · 5 captions

→

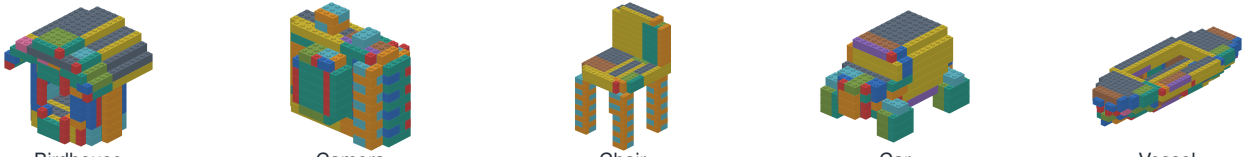
LoRA training
Qwen3-1.7B

→

Text-to-brick pipeline
Generate + repair

(c) LLM-assisted data distillation

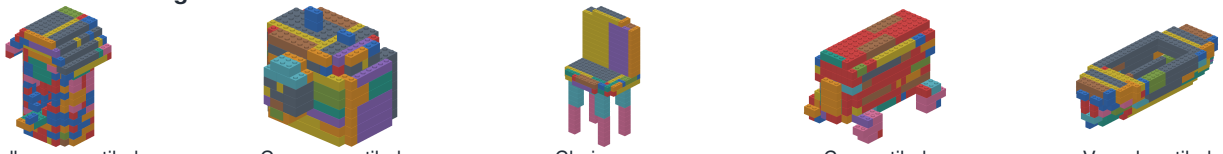
Constructed dataset examples



Birdhouse | Camera | Chair | Car | Vessel

(d) BrickMore model generation

Qwen3-1.7B outputs



Birdhouse · retiled | Camera · retiled | Chair · raw | Car · retiled | Vessel · retiled